

Mock EXAM (2)

Time Allowed: 3 Hours

Course : Introduction Data Science for Business

Instructor(s) : Dr. Ahmed ATIL.

STUDENTS INSTRUCTIONS	
ANSWER	Answers must be in English.
CALCULATOR	Students need / are allowed to use a own LAPTOP
DICTIONARY	
DOCUMENTATION	Documentation allowed. Computer allowed: Students are allowed to use their own documentation as well as their laptop computers and all the information contained. Students are not allowed to share notes.
APPENDIX	Cereals .xlsx Sheet provided.

100 MARKS IN TOTAL, AWARDED ACCORDING TO THE FOLLOWING TABLE:

Question 1	25 points
Question 2	25 points
Question 3	25 points
Question 4	25 points
<i>Total</i>	<i>100 points</i>

Mock EXAM

Introduction Data Science for Business

Healthy Breakfast

Data taken from Carnegie Mellon's DASL

This datafile contains nutritional information and grocery shelf location for 77 breakfast cereals. The data states that adults should consume no more than 30% of their calories in the form of fat (women) or 63 grams (men) of protein daily, and should provide for the remainder of their diet with complex carbohydrates. One gram of fat contains 9 calories and carbohydrates and protein. A "good" diet should also contain 20-35 grams of dietary fiber.

Datafile Name: Cereals

Datafile Subjects: [Food](#) , [Health](#)

Story Names: [Healthy Breakfast](#)

Reference: Data available at many grocery stores

Authorization: free use

Description: Data on several variable of different brands of cereal.

A value of -1 for nutrients indicates a missing observation.

Number of cases: 77

Variable Names:

1. Name: Name of cereal
2. mfr: Manufacturer of cereal where A = American Home Food Products; G = General Mills; K = Kelloggs; N = Nabisco; P = Post; Q = Quaker
3. type: cold or hot
4. calories: calories per serving
5. protein: grams of protein
6. fat: grams of fat
7. sodium: milligrams of sodium
8. fiber: grams of dietary fiber
9. carbo: grams of complex carbohydrates

Question 1 (25 points) Import & Data Manipulation

a) Importing: (2 points)

- Import Cereals .xlsx to Jupyter Notebook (df)
- Write a code to display a summary of the basic information about a specified DataFrame and its data.
 - ✓ data-frame columns, data-frame Dimensions, Data types, Data Information, data etc ...

b) Cleaning Data: (8 points)

- Clean data-frame (df) –
 - **Quantitative variables:** Replace by the Mean: *Calculates the column mean by MFR use the mean as the replacement value for each missing value in the column.*
 - **Categorical variables:** Replace by the Mode : *Calculates the column mode and uses the mode as the replacement value for each missing value in the column.*

c) Data Manipulation: (15 points)

- From given data frame (df) print first and last six rows
- Find the Minimum "calories" cereal name.
- Print the list of cereal name with Sugars rate = 0
- Print the list of cereal name with the following four (4) conditions:
 - Calories rate > mean and Protein rate > mean
 - Carbo rate < mean and Potass rate < mean
- Print the Minimum of "Vitamins_Rate & sodium_Rate" by Cereal Type
- Print the Mean of "Vitamins_Rate & sodium_Rate" by (Cereal Type & mfr)
- Using a loop (for ... in ...) to select a column name from a list of columns:
 - ✓ Print the list of cereal names with Min and the Max of each column.

Question 2 (25 points) Data Visualization

For the Data Visualization, we are using Pandas and Matplotlib to visualize (df) Data.

Using Matplotlib, draw the graphs with ("Title", "Label-Name", " Legend" , Color, etc....)

- The graph of the mean of ('calories' and 'potass') by 'mfr' .
- Read 'sugars' & 'calories' data and show it using a scatter plot
- In Horizontal bar chart display the max of 'Protein ' per 'mfr'.
- The graph of the mean of ('fat', 'fiber', 'cups', 'weight') by 'type'.
- In Pie chart display the maximum of 'sugars ' per 'mfr',
- Read the maximum of 'sugars ' of each 'mfr and show it using the histogram to see most common ranges.

Question 3 (25 points) Programming Algorithm challenges

The manager data scientist asks you to publish the **List of Cereals name** by **Nutrition Category** according to the following rules:

1) Nutrition Category = (1 , 2 or 3)

= 1, if the Global Nutrition Index > 2

= 2, if the Global Nutrition Index < 2 & > 0,5

= 3, if the Global Nutrition Index < 0,5

2) The Global Nutrition Index = Positive Nutrition / Negative Nutrition

Positive Nutrition Index = ('calories' + 'protein' + 'vitamins')

Negative Nutrition Index = (1+ 'sodium' + 'sugars' + 'fat')

Question 4 (25 points) Machine Learning

You have been appointed as a data scientist at **"AZER-FOOD"** a global leader in Genetically Modified Organisms (GMO) products, particularly in the development of new cereals. The head of new cereal development has provided you with a dataset of 20 new cereals, which is referred to as **"New_Cereal_List.xls"**

Your task is to use your machine learning skills to build a program that categorizes these 20 cereals into one of three categories: **Cat1, Cat2, or Cat3**. These categories should be based on the characteristics and features of the cereals, as determined by the results of the cereal project. In (Q1, Q2 and Q3)

The manager data scientist asks you to publish the **List of New Cereals by Nutrition Category**.

